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High impedance fault detection device based on edge artificial intelligence☆

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ABSTRACT

The current high impedance fault (HIF) detection devices are mainly based on threshold method, the hardware cannot meet the computing requirements of artificial intelligence (AI), and the software structure is difficult to adapt to the upgrade requirements of AI. These AI detection algorithms are mainly based on simulation analysis and laboratory testing and have not yet been applied practically. The paper develops an edge AI-based HIF detection device, it combines the main body and independent AI module. The AI module embeds an edge identification algorithm and shares the sampling data with the main body. The detection result is achieved from intelligent classifier in edge AI module and fed back to the main device. Finally, the device is tested on the Luohe distribution network true test platform, the detection accuracy for multiple types of fault is 98.6%.

1. Introduction

Due to the weak fault characteristic signal, the HIF of the distribution network is difficult to be accurately identified and handled. The fault is likely to cause personal electric shock accidents, and the fault arc ignites surrounding flammables to expand fire risk (Guo, 2017). The effective identification of HIF has received more attention, it can be regarded as a pattern detection task, using signal eigenvalues to distinguish HIF phenomena from non-HIF (Silva et al., 2018).

Domestic and foreign scholars have applied artificial intelligence technology to detect HIF (Ledesma, 2020). used two-level ANN to identify high-impedance fault based on synchronized measurement signal, the method considers load variation, the presence of DG, and network reconfigurations (Gomes et al., 2018). defines standard deviation, and Shannon's entropy of high-frequency current as fault feature, and use filed fault samples to train boosted decision trees (Sarwar et al., 2020). employs binary and multiclass support vector machine to detect all types of HIF, the method is also robust against capacitor and load switching transients in the distribution network (Chakraborty and Das, 2019). calculates even harmonic distortion index of voltage waveforms to differentiates HIF from voltage sag-swells, capacitor/load switching, transformer and feeder energization, power electronic loads, arc furnace loads, and DGs (Veerapandiyan et al., 2020). defines absolute sum of the square of details coefficient as detection feature based on discrete wavelet transform analysis, uses fuzzy inference system classifier to discriminate the HIF from other conventional faults based on graphical language (Sekar and Mohanty, 2020). adopts morphology gradient filter to extract the statistical features and develops fuzzy membership functions based on random forest rule set to discover HIF (Soheili and Sadeh, 2017). create basic attributes set including ratio ground, odd and even harmonics, wavelet transform, combinational attempt of harmonic behavior, mathematical morphology, proposes evidential reasoning method to detect HIF in a wider detection scope and higher detection rates (Zhang et al., 2020). integrates historical data from multiple distribution networks to enrich HIF faults samples and detect HIF with a basic cloud convolutional neural network model (Li et al., 2020). obtains the declining periodic components of zero-sequence current to analyze phase difference and fault resistance.

The work proposed the AI algorithm embedded scheme to structural issue hyindering AI application. The device combines the main body and independent AI module, it can realize the complete independence between main body and AI module, the two parts can be upgraded or even

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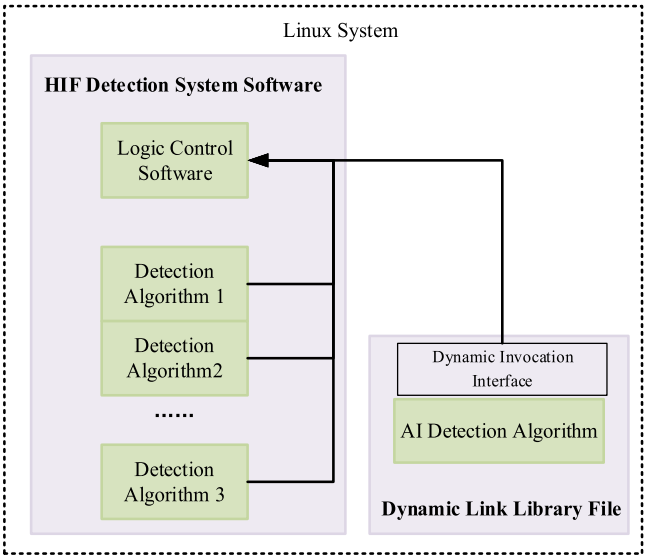


Fig. 1. Schematic diagram of dynamic link library solution.

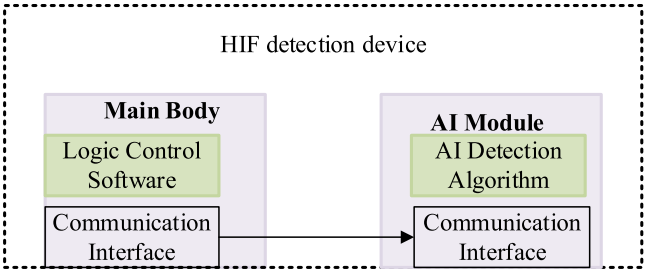


Fig. 2. Schematic diagram of independent algorithm module solution.

replaced at will. The device is tested on the Luohe true test platform, the detection accuracy with 98% shows that it met the needs of field operation.

2. The design scheme of the detection device

2.1. AI algorithm embedded scheme

The current distribution network fault identification and protection device only needs to use DSP+ARM dual processors, DSP is used to process fault transient data, ARM is used for control logic judgment, the algorithm program is relatively simple, and the embedded application of AI algorithm has not yet been realized. Due to the black-box properties and early instability of AI algorithms, flexible upgrade requirements and protection of AI algorithms need to be considered.

2.1.1. Scheme 1: dynamic link library

The scheme principle of the dynamic algorithm library is shown in Fig. 1. The AI algorithm is independently designed and formed a dynamic link library. The dynamic invocation interface is set in the main program of the HIF detection device to directly call the AI algorithm to detect HIF based on transient voltage and transient current data processed by DSP. The main program and the AI algorithm library are relatively independent. If offline trained AI algorithm is upgraded, The AI algorithm library file needs to be updated through the Ethernet port or USB port, and the main program also needs to be recompiled. When the main program needs to be upgraded, it will not affect the AI algorithm. This solution requires that the dynamic library of the algorithm must be embedded in the software of the main device, and there is a risk of program necrosis.

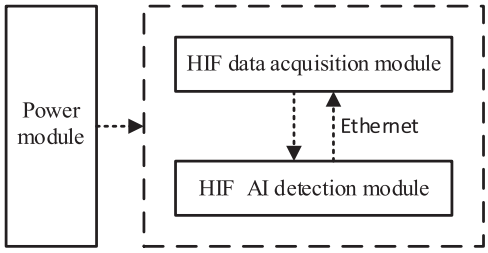


Fig. 3. Structure diagram of HIF detection device.

2.1.2. Scheme 2: independent algorithm module

The principle of the independent algorithm module solution is shown in Fig. 2. The AI algorithm is developed on an independent board, which shares the sampling data with the main body of the HIF detection device, and the AI module calculates and feeds back the results to the main device. This solution can realize the complete independence between main body and AI module, the two parts can be upgraded or even replaced at will. The recorded wave data after the single-phase ground fault and detection result are transmitted between the two parts via Ethernet. If each 20 cycles of fault data have few hundred kB volume, and for 1000 M Ethernet transmission, the delay is less than 10 ms. The calculation result is only dozens of bytes, and the influence of transmission delay can be ignored.

2.1.3. Final solution

In the independent algorithm module solution, there is no hardware coupling between the AI module and the main device, they communicate information according to the standard protocol, and the AI module can realize plug-and-play. The AI learning needs to go through three stages including (1) abundant data, lack of knowledge; (2) moderate data, partial knowledge; (3) scarce data, and complete knowledge, different HIF detection methods for distribution network based on supervised learning, guided learning and evolutionary learning are proposed for different intelligence levels (Hao, 2020). The first stage is suitable for initial research, based on complete simulation data, traversing all fault environment factors. The second stage is suitable for experimental exploration. Based on filed data, simulation data and the summarized knowledge in the first stage, carrying out the fault identification of guided learning. The third stage is suitable for the production and application stage. Based on a large amount of training and relevant knowledge learning, making itself learning and evolution so that achieve the goals of identification of HIFs adaptively by inputting the relevant topology structure and network parameters of the distribution network and dynamically adjusting the relevant parameters of the algorithm.

At the current stage, supervised learning algorithm is adopted, and the AI algorithm needs to be upgraded and debugged several times. Considering the convenience of upgrading and debugging, the independent algorithm module scheme is obviously the optimal choice on the premise that the performance of other functions is not affected.

2.2. Data interaction solution

The data interaction solution is shown in Fig. 3.

(1) The device hardware is based on ARM+DSP+FPGA mode, and the operating system is Linux;
(2) The AI detection module adopts the Raspberry Pi CPU(BCM2837) as core-board, which is an ARMv8 architecture with a main frequency of 2 GHz. It is equipped with 1 GB RAM to meet the needs of the algorithm for content.
(3) The AI module realizes information exchange through high-speed Ethernet.

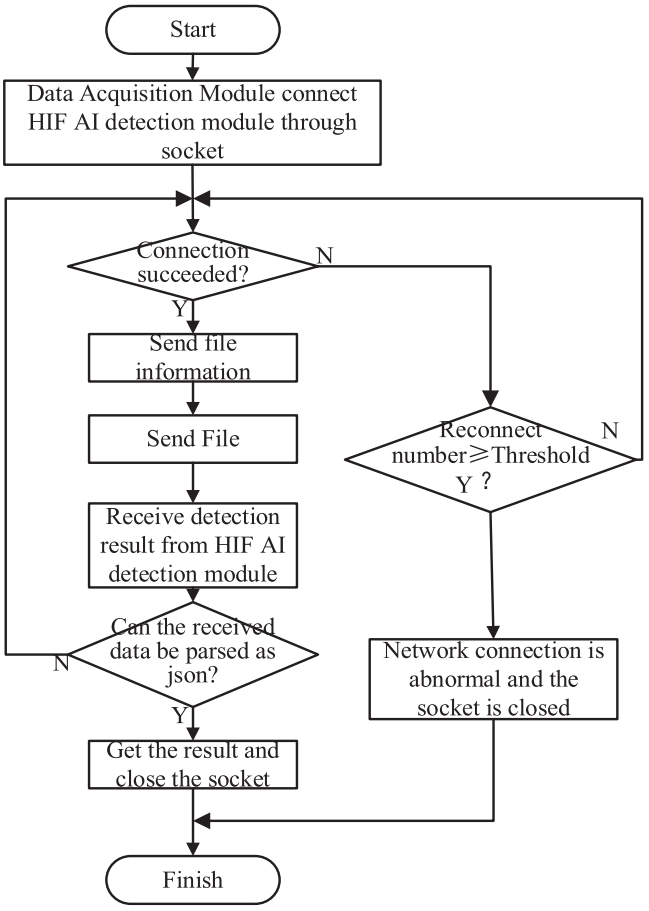


Fig. 4. The work process of AI algorithm module interactive.

(4) The device sends the sampling data from the data acquisition module to the HIF AI algorithm module in real time in the form of wave recording files. The wave recording files are in COMTRADE99 format (including *cfg* files and *dat* files), at least including the zero sequence voltage.
(5) The HIF AI module performs AI calculation after receiving the recorded wave file, and the judgment results are fed back to the HIF device in JSON string format.

2.3. Work process

(1) Create a socket, connect the IP and port, wait for the connection of the HIF data acquisition module, and send a successful connection prompt to the HIF data acquisition module after the connection is successfully established.
(2) Receive the file information sent by the HIF data acquisition module and obtain the size and md5 of the *cfg* file and *dat* file.
(3) Receive the file. When the file is large, the HIF data acquisition module may trigger a timeout due to the delay caused by sub-packet receiving. Therefore, every time a packet is received, an ACK frame is sent to the high-impedance ground fault data acquisition module. After receiving, perform md5 verification, and return the result to the HIF data acquisition module
(4) Parse the recording file and run the algorithm to get the result. After the result is obtained, the information of the Result type is returned to the HIF data acquisition module.
(5) The main device and the AI module work together, and the process can be divided into the following three links:

- The main device performs real-time detection to preliminarily determine whether there is a suspected HIF event in the feeder.
- If there is a suspected HIF event, send the wave-recording file to the AI module.
- The AI module receives the wave recording file and parses it to obtain the waveform data. According to the waveform data, run the algorithm to determine whether the suspected event is a HIF and send the result back to the main device. If it is a HIF, the main device will emit an alarm.

(6) Exception handling. When the following exceptions are encountered, it is considered that manual resolution is required, an error is reported, and the program is exited:

- The file that the high resistance ground fault data acquisition module is trying to send does not exist.
- The wave recording file cannot be parsed.
- The number of disconnection and reconnection times reaches the maximum value (Fig. 4).

3. The edge AI detection algorithm

This device adopts the HIF detection method of distribution network based on supervised learning (Bai, 2022). Firstly, the three-phase voltage and zero-sequence voltage waveforms of the bus are collected, and the waveforms of the first half cycle before the fault and one-half cycle after the fault are selected, and the time-frequency matrix is constructed by using the local eigenvalue decomposition algorithm and fed to convolutional neural network The detection of the HIF in the distribution network is realized through the trained convolutional neural network (Fig. 5).

The specific steps of HIF detection in a convolutional neural network are as follows:

(1) Obtain and organize training samples and test samples
The software simulation model and the physical experiment platform are used to simulate various situations, and the

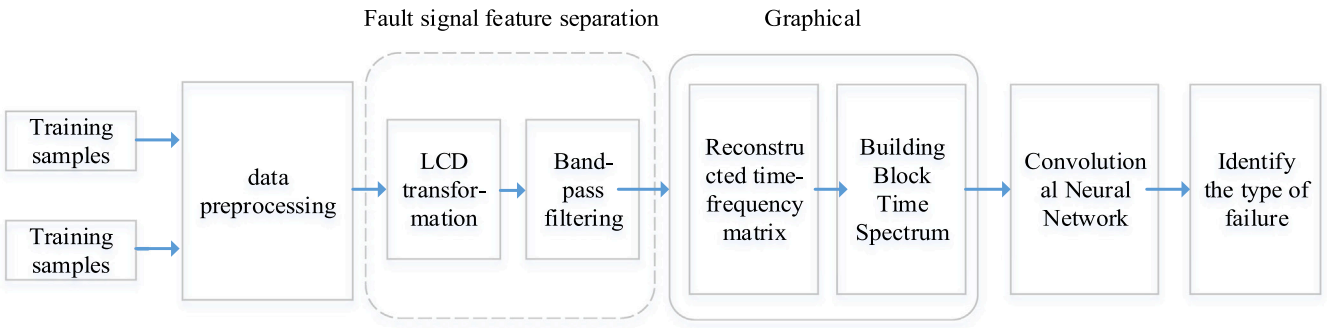


Fig. 5. Diagram of fault type identification.

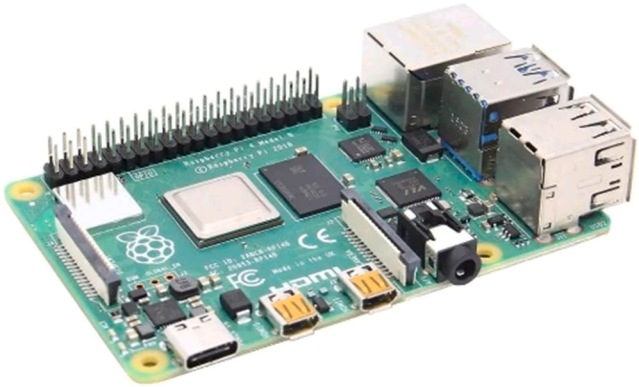


Fig. 6. AI detection module.



Fig. 7. HIF detection device based on edge AI.

waveforms under each working condition are collected by the wave recording device, which are divided into training samples and test samples.

(2) (2) Data processing

The sampling frequency is set to 10 kHz, and the simulation length is selected to be 0.5 s. Intercept the first half-cycle of the fault and one and a half cycles after the fault with a total length of 0.04 s, that is, the three-phase voltage and zero-sequence voltage data of 400 points.

(3) (3) Signal time-frequency decomposition

The Hilbert time spectrum is obtained by local characteristic-scale decomposition and transformation and is reconstructed into a time-frequency matrix of 5 frequency bands by the band-pass filtering algorithm.

(4) (4) Graphical time-frequency matrix

The reconstructed time-frequency matrix is constructed into block time-frequency matrices and arranged from top to bottom in the order of three-phase voltage and zero-sequence voltage to form a 20 × 40 block time-frequency spectrum.

(5) CNN training

The training samples are sorted by type and the corresponding labels are established: 1) HIF; 2) general single-phase ground fault; 3). Transient disturbance conditions. The 7-layer CNN is trained by the backpropagation algorithm and the CNN classification model is obtained.

(6) Identification of the fault type

The test samples are labeled according to their labels, and the trained CNN model is used to classify the test samples.

4. The HIF detection device

A HIF detection module scheme based on the Raspberry Pi distribution network is designed. The module has a built-in evolutionary learning AI identification algorithm, which supports offline upgrade and online learning upgrade. The hardware used in the HIF intelligent identification module is Raspberry Pi 4B (Jabbar, 2021). Raspberry Pi 4B is an ARM-based microcomputer motherboard that runs the Linux-based Raspberry Pi OS system and has the basic functions of a PC (Fig. 6).

The device includes a main board, a digital input module, a digital output module, an analog signal module, a power supply module, a communication module, an AI module, a human-computer interaction module, and the like. The HIF detection device is shown in Fig. 7.

5. Device true test

Luohe distribution network true test platform has 6 lines, including 2 real overhead lines, 1 real cable line, and 3 simulated lines. The incoming line of the real test site is a dedicated line, the transformer capacity is 3150 kVA, and a step-up transformer, a step-down transformer and a voltage regulator are installed. The grid structure of the real test field is relatively simple, and there is a connection between the lines. The length of the two real overhead lines is 5 km, and the length of the real cable line is 3 km, all of which are no-load lines; the other three



Fig. 8. HIF detection device in true test.

Table 1
Test items and test conclusions.

Grounding Mode	Test Items	Test Times	Correct Times	Accuracy
The neutral point is not grounded	Metallic single-phase ground	4	4	100%
	1000 Ω single-phase ground	4	4	83.33%
	2000 Ω single-phase ground	6	5	100%
	Instantaneous arc single-phase ground	3	3	100%
	Cable arc single-phase ground	9	9	100%
Small resistance	250 Ω single-phase ground	3	3	100%
	Cable arc single-phase ground	4	4	100%
Arc suppression coil	Metallic single-phase ground	6	6	100%
	500 Ω single-phase ground	6	6	100%
	1000 Ω single-phase ground	9	9	100%
	2000 Ω single-phase ground	10	10	100%
	Instantaneous arc single-phase ground	3	3	100%
	Cable arc single-phase ground	6	6	100%
Total		73	72	98.6%

are analog lines, and there are 4 branch lines. The analog circuit is simulated by impedance, adopts a T-shaped structure, and concentrates parameters. The system is set up with a low-voltage station area and configured with a 315 kVA station transformer. The system is equipped with a motor load of 400 kW and a harmonic impact load of 100–200 kW. The test field is equipped with 2 fault areas, which can realize the flexible access of the equipment under test. The system is equipped with two sets of grounding devices. The grounding methods include ungrounded, arc suppression coil grounding, small resistance grounding, medium resistance grounding, dynamic grounding and standby interface. Two sets of arc suppression coil grounding devices are provided, and the grounding resistance can be dynamically adjusted in the range of 0–32 kΩ (Fig. 8).

To verify the reliability of the HIF detection device in the actual distribution network, a trial operation test was carried out for. The true test considers three grounding condition such as the ungrounded, small resistance grounding and arc suppression coil grounding. The test set 73 faults including 250 Ω transition resistance, 1000 Ω transition resistance, and 2000 Ω transition resistance, instantaneous arc, cable arc and other ground faults. The developed device can detection 72 faults, the correct rate is 98.6%, test details are shown in the Table 1.

6. Conclusion

(1) This paper proposes an independent algorithm module scheme. The AI algorithm is developed based on an independent board and shares the sampling data with the main body of the high resistance identification device. After the AI module calculates, the results are fed back to the main device.
(2) This paper develops an AI module based on Raspberry Pi, and embeds a supervised detection algorithm.
(3) The paper develops an edge AI-based HIF detection device, and the fault identification test is carried out in the real test field of the distribution network, and the correct rate of identification line selection is 98.6%.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

The authors do not have permission to share data.

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